How sensitive are finite RH criteria? Certified Davenport–Heilbronn controls

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Abstract

Finite positivity tests related to the Riemann Hypothesis require calibration against functions with known off-line zeros. We study the classical Davenport–Heilbronn function, additional character combinations, and two Epstein zeta functions. An argument-principle computation certifies the complete Davenport–Heilbronn zero atlas through height 340: 235 critical-line zeros and seven reflected off-line pairs in the upper half-plane. A leading-term Dirichlet-series bound proves that the zero at height 85.69934849 has the largest folded angular deviation among all zeros. Schoenberg's sufficient sector theorem consequently gives PF_{435} for the ordinary folded Taylor coefficients. Nevertheless, all 18,291 contiguous minors at ranks 430–520 and shifts 0–200 are positive at 32,768-bit precision. The isolated quadratic factor first has a negative shift-one minor at rank 436. An elementary masking lemma explains why such a factor need not force failure in a fixed-shift ladder; a dominant-pole argument proves eventual positivity of the full Davenport–Heilbronn shift-one ladder. We also give certified negative geometric-form trials for seven controls, including a conductor-17 test of a prediction recorded before its new window search, and finite-zero-sum attribution diagnostics. The window data are negative-support upper bounds, not certified minimum supports. Neither the fitted calibration nor the finite Toeplitz grid yields a universal zero-free region or a proof of the Riemann Hypothesis.

Keywords: Davenport–Heilbronn function; Weil quadratic form; Toeplitz minors; total positivity; interval arithmetic; negative controls; zero detection.
MSC 2020: 11M26, 11M41, 15B05.

1 Introduction

An all-orders positivity criterion can characterize the Riemann Hypothesis without making a finite instance a sensitive detector of an off-line zero. The distinction matters when choosing a finite computational target. Davenport–Heilbronn functions supply controls with a functional equation and zeros off the symmetry line. Their classical theory and numerical zero computations are established background [1, 2, 3]. All seven off-line centers in Table 1 were already reported: the first four by Spira and the final three by Balanzario–Sánchez-Ortiz, as recorded in [2]. Our contribution is the complete interval count and global angular comparison, rather than discovery of these zero locations.

Our main result is a reproducible calibration for one specified family of Toeplitz minors. We certify the steepest folded Davenport–Heilbronn zero and the resulting PF_{435} sufficient guarantee, then test the requested grid beyond the isolated quadratic's first negative minor. Every tested full-function minor is positive. This is a failure of sharpness for the specified solid grid. It does not locate the globally first negative minor, and does not prove PF_{520}.

The window experiments address a different finite criterion. We construct rational trial functions with negative Weil-type quadratic form directly from the geometric side. The existing `teal-sea/zeta-lab` report [6] already found the classical Davenport–Heilbronn negative window near $C = 31$; that observation is prior work. We refine the finite-family search, add other controls, and distinguish the resulting upper bounds from a theorem about the minimum negative support. Bombieri's fixed-support spectral analysis [4] is direct prior art for this finite-to-infinite distinction.

The contributions of this draft are:

- a complete interval-certified Davenport–Heilbronn atlas through height 340 and a global steepest-angle reduction;
- a fully resolved 18,291-cell solid-minor grid, with a second determinant identity checking selected high-rank cells;
- a common-degree seven-control window calibration, a recorded conductor-17 prediction test, and enclosed finite-atlas quartet contributions;
- an elementary fixed-shift masking lemma and a direct eventual-positivity argument for the classical Davenport–Heilbronn shift-one ladder.

These are calibration results. The analytic reductions use classical functional equations, factorization and sector theorems; the numerical enclosures use Arb. They have not undergone independent external review.

2 Normalizations and the finite tests

2.1 Character combinations and Epstein controls

Let χ be a primitive odd nonreal character of conductor q, and put

$$\epsilon_\chi = \frac{\tau(\chi)}{i\sqrt{q}}, \qquad \kappa_\chi = \frac{\operatorname{Im}\epsilon_\chi}{1+\operatorname{Re}\epsilon_\chi}, \qquad H_\chi(s) = \frac{(1-i\kappa_\chi)L(s,\chi)+(1+i\kappa_\chi)L(s,\bar\chi)}{2}. \tag{1}$$

We exclude the singular normalization $1+\operatorname{Re}\epsilon_\chi = 0$. The Dirichlet coefficients are real, periodic, and have first coefficient one. The completion

$$\Lambda_\chi(s) = (q/\pi)^{s/2}\Gamma((s+1)/2)H_\chi(s) \tag{2}$$

is entire and satisfies $\Lambda_\chi(s) = \Lambda_\chi(1-s)$. The classical control has $q = 5$, $\chi(2) = i$, and period $(0, 1, \kappa, -\kappa, -1)$, where

$$\kappa = \frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}. \tag{3}$$

We also use $H_d(s) = E(s, x^2+dy^2)/2$, $d = 5, 6$, normalized by counting representations. The class-number-two decompositions are

$$H_5(s) = \tfrac{1}{2}\{\zeta(s)L(s,\chi_{-20}) + L(s,\chi_{-4})L(s,\chi_5)\}, \tag{4}$$

$$H_6(s) = \tfrac{1}{2}\{\zeta(s)L(s,\chi_{-24}) + L(s,\chi_{-3})L(s,\chi_8)\}. \tag{5}$$

Here χ_D is the primitive quadratic character of fundamental discriminant D. The completion has gamma factor $(\sqrt{d}/\pi)^s\Gamma(s)$ and poles at 0 and 1. The entire symmetric target is therefore $s(s-1)(\sqrt{d}/\pi)^s\Gamma(s)H_d(s)$. The contour computation uses $F_d(s) = (s-1)H_d(s)$, removing the pole at 1; the prefactor relating it to the entire target introduces no additional zero inside the contours used here. Genus decompositions and their normalizations are checked against independent representation counts. The background for these controls is discussed in [3].

2.2 Ordinary coefficients and solid minors

For the classical completion define the folded entire function

$$\frac{\Lambda_5(1/2+z)}{\Lambda_5(1/2)} = F(z^2), \qquad F(w) = \sum_{n\ge 0} \mu_n w^n, \qquad \mu_0 = 1. \tag{6}$$

All Toeplitz tests use the ordinary coefficients μ_n. The classical Jensen coefficients $n!\mu_n$ are different data and are not substituted into these matrices. With $\mu_n = 0$ for $n < 0$, write

$$D_{r,s} = \det(\mu_{s+i-j})_{0\le i,j<r}, \qquad D_{0,s} = D_{r,0} = 1. \tag{7}$$

PF_m means nonnegativity of all Toeplitz minors of order at most m, including arbitrary row and column sets. Our solid grid is only a subset.

2.3 Quadratic forms and support

We use physical support $[-S/2, S/2]$, with

$$S = \log C. \tag{8}$$

For a real trial function let $\mathcal{F}_f(z) = \int_{-S/2}^{S/2} f(x)e^{zx}\,dx$. The spectral form associated with a real symmetric completion is

$$Q_H(f) = \sum_{\rho} \mathcal{F}_f(\rho - \tfrac{1}{2})\overline{\mathcal{F}_f(\tfrac{1}{2} - \bar{\rho})}. \tag{9}$$

Its geometric form uses the coefficients in $-H'/H = \sum_{n\ge 2} c_n n^{-s}$. For a character combination these coefficients need not have prime-power support or one sign. All active integers $n < C$ are included. The odd gamma factor and conductor term in (2) are used, and there is no pole block. The Epstein forms have their own gamma and pole terms. The gamma duplication identity is used before reusing entry formulas.

The finite trial family consists of normalized cosines through order N in the even sector and normalized sines through order N in the odd sector, together with the constant even mode. Each sector is orthonormal in $L^2[-S/2, S/2]$. Stored coordinates use the shifted exponentials $S^{-1/2}e^{2\pi i n(x+S/2)/S}$, so the real centered sine and cosine modes include the phase $(-1)^n$. This phase is retained in zero-side evaluation. An interval LDL factorization proposes a rational vector at its first negative leading pivot. The full geometric Rayleigh quotient of that exact vector is then enclosed independently of the pivot sign. A negative quotient proves a negative full-form trial. Positivity of the finite matrix proves only positivity on that finite space. The spaces obtained by changing S are not nested, so the searches below do not provide monotone brackets for the first negative full support.

3 The complete classical Davenport–Heilbronn atlas

Lemma 3.1 (Positive theta kernel)**.** *The classical completion is positive on the real axis. Every ordinary folded coefficient μ_n is strictly positive.*

Proof. Set $\theta(t) = \sum_{n\ge 1} n a_n e^{-\pi n^2 t/5}$. Poisson summation for the odd character combination gives $\theta(t) = t^{-3/2}\theta(1/t)$. For $t \ge 1$, put $v = \pi t/5 > 1/2$. Since $a_2 = \kappa > 0$ and $|a_n| \le 1$,

$$\theta(t) \ge e^{-v} - \sum_{n\ge 3} n e^{-vn^2} \ge e^{-v} - \frac{e^{-4v}}{2v} > 0. \tag{10}$$

The middle inequality uses the decreasing function xe^{-vx^2} on $[2,\infty)$. Consequently $\Psi(u) = e^{3u/2}\theta(e^{2u})$ is even and positive. Mellin inversion gives

$$\Lambda_5(1/2+z) = 4\sqrt{\pi/5}\int_0^\infty \Psi(u)\cosh(zu)\,du. \tag{11}$$

The kernel decays faster than any exponential. This proves real-axis positivity and positivity of every even Taylor moment. □

Theorem 3.2 (Computer-assisted atlas and steepest angle)**.** *The completed classical Davenport–Heilbronn function has precisely 249 zeros, counted with multiplicity, in the upper half-plane through height 340. They consist of 235 critical-line zeros and seven reflected off-line pairs. All are simple. Every completed zero satisfies* $|\operatorname{Re}\rho - \frac{1}{2}| < 1$. *The zero near* $0.8085171825 + 85.69934849i$ *has the largest folded angular deviation among all completed zeros.*

Proof with explicit numerical inputs. The first coefficient of H_5 is one, and Arb gives

$$\sum_{n\geq 2}|a_n|n^{-3/2} = 0.7375906494\ldots < 1. \tag{12}$$

The sum decreases with the real part. Thus H_5 has no zero at $\operatorname{Re} s \geq 3/2$; the completed functional equation gives the reflected exclusion at $\operatorname{Re} s \leq -1/2$. There are no gamma poles inside this strip.

On the rectangle $[-1/2, 3/2] + i[0, 340]$, the real bottom has zero argument increment by Lemma 3.1; its gamma factor is positive there. The other three sides are covered by 5,208 Taylor image rectangles excluding zero. A convex zero-excluding image gives a continuous argument increment of magnitude less than π, so principal endpoint increments may be summed. Their total encloses $2\pi \cdot 249$. There are 235 disjoint critical-line sign brackets and seven disjoint simple right-hand Rouché disks. Functional-equation reflection supplies seven left-hand disks. Their multiplicity count exhausts 249, proving completeness and simplicity. The atlas is replayed at higher precision with additional subdivision where needed.

For $\rho = 1/2 + b + i\gamma$, the zero of F at $w = (\rho - 1/2)^2$ has angular deviation

$$\phi_\rho = 2\arctan\frac{|b|}{|\gamma|} \tag{13}$$

from the negative axis. The height-85.6993 disk has $\phi = 0.0071999578790\ldots$, greater than the deviations of all other off-line disks and greater than $2\arctan(1/340)$. The strip exclusion therefore controls every higher zero as well. Every comparison uses enclosing balls, rather than the rounded values in Table 1. □

4 The sector guarantee and the high-rank grid

Theorem 4.1 (Sufficient sector guarantee)**.** *The ordinary coefficient sequence in* (6) *belongs to* PF_{435}.

Proof. The folded function has order 1/2, hence a genus-zero product with constant one. By Theorem 3.2 every zero lies in the wedge about the negative axis of half-angle ϕ, where interval calculation gives

$$436 < \frac{\pi}{\phi} = 436.3348656172\ldots < 437. \tag{14}$$

Table 1: Right-hand classical Davenport–Heilbronn zeros below height 340. The reflected zeros and 235 critical-line zeros exhaust the contour count. Rounded coordinates are not certificate inputs.

$\operatorname{Re}\rho$	$\operatorname{Im}\rho$
0.8085171825	85.69934849
0.6508300806	114.1633427
0.5743560505	166.4793059
0.7242576946	176.7024612
0.8695305796	240.4046724
0.8195495921	320.8764897
0.7682231236	331.0502594

Schoenberg's sufficient sector theorem, stated as Theorem B in [5], gives PF_{435} for every finite conjugation-invariant product prefix: its wedge half-angle is less than $\pi/436$. The prefixes converge locally uniformly. Each fixed Toeplitz minor is a polynomial in finitely many coefficients, so nonnegativity passes to the entire function. □

For the isolated quadratic $A(w) = 1 + 2\cos\phi\, w + w^2$, expansion of the tridiagonal determinant gives

$$D_{r,1}(A) = U_r(\cos\phi) = \frac{\sin((r+1)\phi)}{\sin\phi}. \tag{15}$$

Equation (14) makes rank 436 its first negative shift-one minor. The angular rank scale is $\pi|\gamma|/(2|b|)$, retaining the offset b.

Proposition 4.2 (Certified solid grid)**.** *For the full folded Davenport–Heilbronn function, $D_{r,s} > 0$ for every $430 \le r \le 520$ and $0 \le s \le 200$.*

Computational verification. We generate 721 ordinary coefficients through index 720 at 32,768 bits. The positive rescaling $\mu_n \mapsto 100^n \mu_n$ preserves determinant signs. Desnanot–Jacobi condensation evaluates

$$D_{r,s} = \frac{D_{r-1,s}^2 - D_{r-1,s-1}D_{r-1,s+1}}{D_{r-2,s}}. \tag{16}$$

Every divisor is required to exclude zero; unresolved entries invalidate their descendants. All 18,291 requested cells resolve positive. This count includes the 91 trivial shift-zero cells. The full sign digest is replayed. For $h_n = [w^n]1/F(-w)$, Jacobi's complementary-minor identity also gives

$$D_{r,s} = \det(h_{r+i-j})_{0\le i,j<s}. \tag{17}$$

It independently checks 24 high-rank cells at ranks 430, 435, 436, 437, 480 and 520, and shifts 1, 2, 5 and 20. These checks share the interval coefficient input and FLINT backend, while bypassing the condensation divisions. □

Figure 1 displays the distinction between the isolated factor and the full-function grid. The sufficient sector bound is not sharp for this tested family. Arbitrary row and column sets and larger shifts remain untested, so Proposition 4.2 supplies no global first-failure rank or PF_{520} claim.

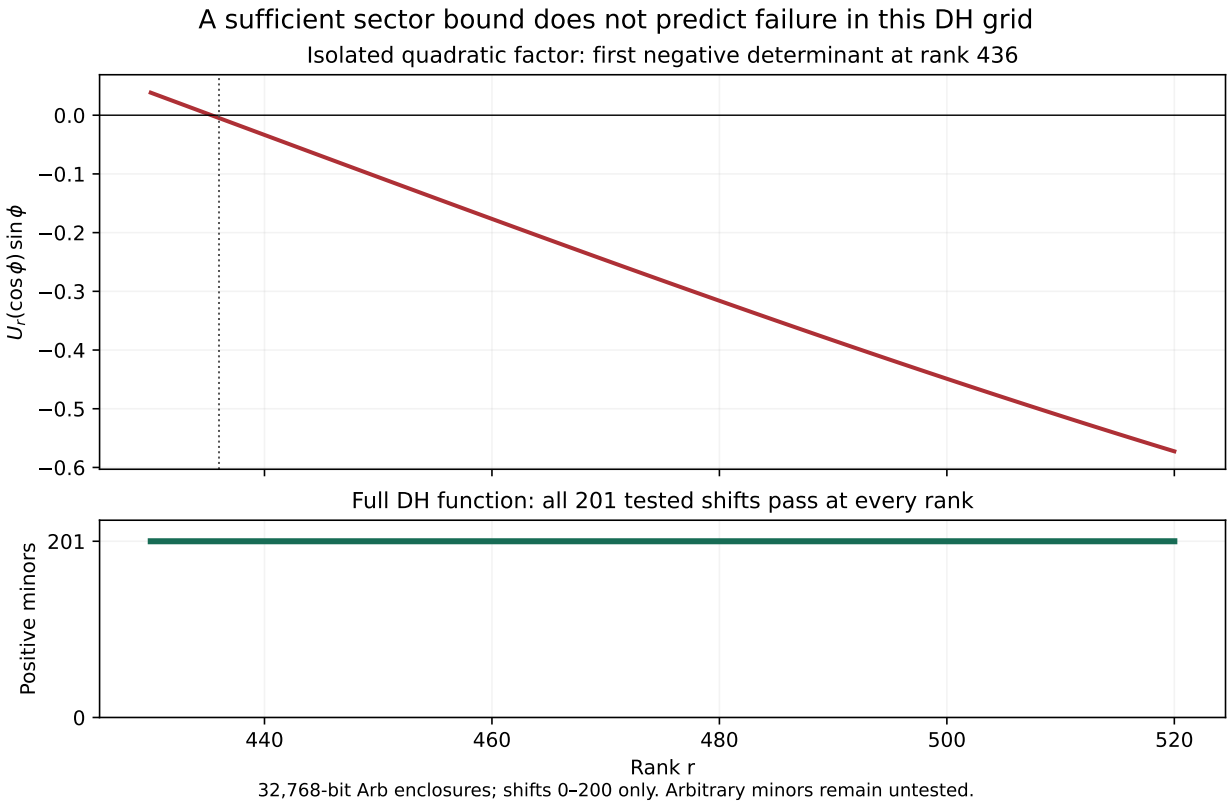


Figure 1: The isolated quadratic determinant, scaled by $\sin\phi$, changes sign at rank 436. Every one of the 201 tested full-function shifts remains positive at each rank through 520. The lower panel counts signs; it does not compare determinant magnitudes or assert positivity of arbitrary minors.

5 Why multiplication can conceal a negative factor

Toeplitz multiplication and Cauchy–Binet expand a product minor into a sum of products of factor minors. One negative term need not determine the sum. The following elementary example makes that obstruction quantitative. It is included as an explanation, without a novelty claim.

Lemma 5.1 (A fixed-shift ladder can miss a conjugate pair forever)**.** *Let* $0 < c < 1$ *and*

$$G(w) = (1 + 4w)(1 + 2cw + w^2). \tag{18}$$

Then G *has two nonreal zeros, yet*

$$D_{r,1}(G) \geq \frac{2}{9}\,4^r > 0 \qquad (r \geq 0). \tag{19}$$

Proof. The two nonreal zeros belong to the quadratic, while $1 + 4w$ adds a negative real zero. The reciprocal generating identity gives

$$D_{r,1}(G) = [w^r]\frac{1}{G(-w)} = \sum_{j=0}^{r} 4^{r-j} U_j(c). \tag{20}$$

Writing $c = \cos\phi$ with $0 < \phi < \pi/2$, the finite exponential sum for $U_j(\cos\phi)$ gives $|U_j(c)| \le j+1$. Since $U_0 = 1$,

$$4^{-r} D_{r,1}(G) \ge 1 - \sum_{j=1}^{\infty} \frac{j+1}{4^j} = 2 - (1 - \tfrac{1}{4})^{-2} = \frac{2}{9}. \tag{21}$$

□

This lemma concerns one fixed-shift ladder. Other minors of G may fail. It does not prove the full Davenport–Heilbronn grid by factorization, since the remaining Davenport–Heilbronn factors include further off-line pairs. It explains why the first negative minor of an isolated quadratic is not an upper bound for the corresponding full-function ladder.

Proposition 5.2 (Eventual positivity of the DH shift-one ladder)**.** *Let γ_1 be the lowest positive Davenport–Heilbronn zero ordinate and put $a = \gamma_1^2$. There is a constant $A > 0$ such that*

$$D_{r,1}(F) = Aa^{-r} + O(64^{-r}) \qquad (r \longrightarrow \infty). \tag{22}$$

In particular, the full-function shift-one minors are strictly positive at every sufficiently large rank, despite the off-line zeros.

Proof. The complete atlas places the first two positive critical-line ordinates in $(81/16, 41/8)$ and $(143/16, 9)$, respectively. There is no off-line zero below height 85. Thus the first folded zero $-a$ is simple and unique in $|w| \le 64$, since $(41/8)^2 < 64 < (143/16)^2$. Write $B(w) = F(-w)$. We have $B(0) = 1$, and B is real and nonzero on $[0, a)$. Its simple first crossing therefore has $B'(a) < 0$. Set $A = -1/(aB'(a)) > 0$. The function

$$\frac{1}{B(w)} - \frac{A}{1 - w/a}$$

is analytic on a neighborhood of the closed disk $|w| \le 64$. Cauchy's coefficient estimate and $D_{r,1}(F) = [w^r]1/B(w)$ prove (22). □

This is ordinary dominant-pole analysis, included without a novelty claim. It does not give an explicit last exceptional rank, prove positivity at every rank, or control other shifts. It shows directly how a nearer real zero can dominate this particular DH ladder.

6 Seven-control window calibration

We compare the finite-family negative endpoints at common degree $N = 192$ and 2,048-bit arithmetic. Table 2 reports the lowest off-line zero from each complete finite atlas, the positive finite endpoint, and the certified negative endpoint. Complete atlases reach height 160 for the additional character combinations and 120 for the Epstein controls. A conductor-11 exploratory combination has no off-line zero below 160 and is excluded from the fitted seven-point sample.

The leading zero densities motivate a normalized resolution coordinate, rather than an exchange theorem:

$$T_*(C) = \begin{cases} 2\pi C/q, & \text{odd character completion of conductor } q, \\ \pi\sqrt{C/d}, & \text{Epstein form } x^2 + dy^2, \end{cases} \qquad r = \frac{T_*(C)}{\gamma}. \tag{23}$$

These expressions equate the leading ordinate density with the exponential type budget $S/(2\pi)$. That calculation does not prove that an off-line zero is detected at $T_*(C)$.

Table 2: Common-degree calibration, $N = 192$. C_+ means positive finite evidence; C_- is a certified negative full-form trial. The values are rounded; exact rational endpoints are retained in the supplement. $r = T_*(C_-)/\gamma$.

Control	γ	b	C_+	C_-	r
DH5	85.699348	0.308517	30.628746	30.630219	0.44914
DH13(2)	31.434205	0.083622	33.750244	33.751953	0.51896
Epstein5	15.668250	0.432970	19.825439	19.827148	0.39928
DH17(3)	105.155287	0.163306	133.738281	133.751953	0.47011
DH7(3)	31.580361	0.535699	13.008621	13.010651	0.36980
DH13(6)	25.608746	0.362174	21.099091	21.101334	0.39825
Epstein6	15.117579	0.303749	31.778137	31.781708	0.47828

Before the new conductor-17 window search, the prediction $C = 140$, with the interval $[126, 154]$, was recorded from the earlier three-control calibration. Its atlas proves that $0.6633061848 + 105.15528688i$ is the lowest off-line zero. The negative endpoint $C = 133.751953125$ is 4.46% below the forecast and falls inside the predicted interval. Degree 80 remains positive at every tested point through 200, whereas degrees 128 and 192 give negative trials at 140. Thus a detector-resolution check is necessary even for the empirical test. At degree 256, the tested section is positive at $C = 133.5$ and a rational witness is negative at $C = 133.73828125$. This modest change from degree 192 supports finite-family resolution stability at this control. None of these positive sections certifies a positive full window.

Figure 2 shows the seven finite-family upper-bound coordinates and descriptive fits. The fitted samples are not independent measurements of minimum support. The two Epstein observations offer only limited information on a degree effect, and the conductor-7 control has a zero to the right of 1. These distinctions limit extrapolation to zeta. The simple fit $r = a + kb$ has root-mean-square residual 0.0183. Adding a degree term reduces it to 0.0149, but the largest absolute leave-one-out error in the implied C rises from 16.3% to 19.6%. Seven support upper bounds therefore do not establish a universal 10%-accurate exchange law.

7 Finite-zero-sum attribution

For a normalized real even witness, an off-line quartet represented by $z = b + i\gamma$ contributes

$$4\,\mathrm{Re}\{\mathcal{F}_f(z)^2\}. \tag{24}$$

For a real odd witness the contribution has the opposite sign. A critical-line pair contributes $2|\mathcal{F}_f(i\gamma)|^2$. We refine the atlas disks to radius 10^{-120}, verify containment in their parent disks or sign brackets, and evaluate these finite contributions with interval arithmetic.

Table 3 identifies the largest enclosed negative quartet contribution for each negative endpoint witness. The supplementary records retain the positive line sum separately from the finite total. This answers which quartet carries the negative term within the atlas. It does not prove that the omitted infinite tail is negligible at the scale of the geometric quotient. The exact geometric certificate proves negativity; the finite zero sum supplies mechanism evidence. A non-smooth Fourier witness does not automatically give the tail decay needed to promote that evidence to an independent full spectral certificate.

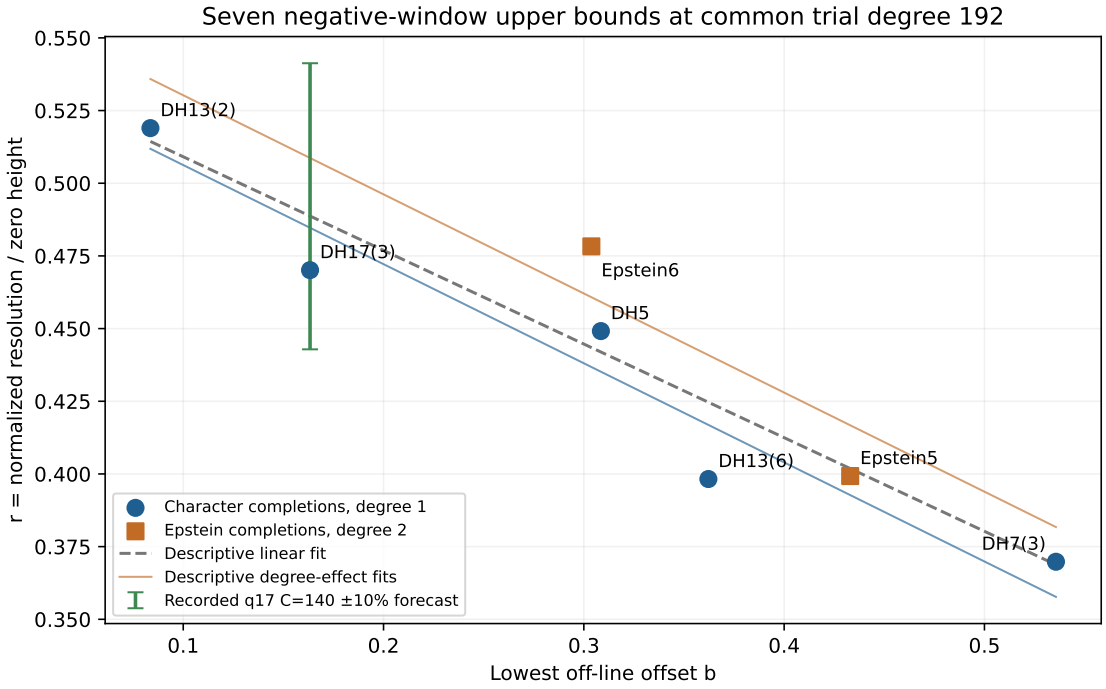


Figure 2: Normalized negative-window upper bounds $r = T_*(C_-)/\gamma$ against the lowest off-line offset b, using common trial degree 192. The fits are descriptive. They do not provide confidence intervals for unknown minimum supports or a zero-free region for zeta. The conductor-17 prediction was supplied before the new window trials.

8 Reproduction and limits of inference

The code uses Python 3.12, python-flint 0.8.0, and Arb/FLINT interval arithmetic. Discovery Newton steps and floating-point pilot scans select candidates only. The accepted root disks, contour images, determinant signs, Rayleigh quotients, and frequency panels are checked with outward enclosures. The supplement records source hashes, input hashes, exact rational trial coefficients, pivot intervals, and replay commands. Regenerable full matrix arrays are omitted from compact records; the initial generator is preserved by hash. Public archival release and external analytic review are pending.

The classical atlas uses `dh_zero_atlas.py` and `replay_dh_atlas.py`. The existing solid grid and complementary checks use `dh_high_rank_scan.py` and `replay_dh_high_rank.py`; no further Toeplitz grid is generated for this paper. New control atlases use `character_zero_atlas.py` and `epstein_zero_atlas.py`. The geometric calibration and finite attribution use `control_window_calibration.py` and `control_zero_attribution.py`.

The finite Schur inertia identity after completion of the square and the actual quadratic Toeplitz determinant formula are checked in Lean 4. Their six-theorem audit uses only `propext`, `Classical.choice`, and `Quot.sound`. The analytic explicit formulas, entire-function factorization, Schoenberg sector theorem, and interval computations remain separate written and computer-assisted dependencies.

The data support testing a calibrated detector. They do not establish *positivity at C excludes every off-line zero below approximately* $12C$. Such an implication additionally requires a uniform theorem over offsets, multiplicities, local zero configurations, and the infinite tail. The present

Table 3: Attribution for each common-degree negative endpoint witness. The dominant negative quartet is the lowest off-line pair in all seven finite atlases. P_H is the sum over the complete atlas below height H; Q is the certified geometric quotient. Their difference is material, so the finite sum is mechanism evidence rather than a negligible-tail spectral certificate.

Control	Parity	Dominant γ	Q	P_H
DH5	even	85.699348	-9.772×10^{-33}	-2.519×10^{-30}
DH13(2)	even	31.434205	-6.173×10^{-17}	-4.418×10^{-13}
Epstein5	even	15.668250	-1.031×10^{-8}	-1.296×10^{-4}
DH17(3)	even	105.155287	-1.483×10^{-43}	-5.005×10^{-40}
DH7(3)	odd	31.580361	-9.555×10^{-10}	-4.135×10^{-6}
DH13(6)	even	25.608746	-3.045×10^{-10}	-8.222×10^{-7}
Epstein6	even	15.117579	-3.789×10^{-9}	-2.180×10^{-5}

offsets are bounded away from zero; their fitted curve does not determine the small-offset limit. Likewise, the solid-grid result does not characterize all minors of a given rank. These controls provide no independent confirmation of a new zeta window-positivity record and no route to RH by themselves.

9 Discussion

Two finite criteria show different sensitivities. The Toeplitz grid remains positive after the isolated factor becomes negative, and Lemma 5.1 demonstrates an exact fixed-shift masking mechanism. Proposition 5.2 proves eventual positivity for the classical Davenport–Heilbronn shift-one ladder itself. The window family produces negative rational witnesses at reproducible cutoffs, with a conductor-17 negative endpoint inside its forecast range. The shared feature is that resolution and the zero configuration must be accounted for before interpreting a finite sign.

The near-term mathematical target is a quantitative detector with a proved tail and cancellation bound. The near-term computational target is a certificate whose full comparison cost is measured, including the infinite complement. Calibration of either target requires controls such as those used here. Neither replaces the missing all-support arithmetic estimate.

A Entry formulas and removable-point contour images

Here are explicit formulas for regenerating the new finite matrices. Put $\omega_m = 2\pi m/S$, $b_k = 2k+2\alpha$, where $\alpha = 1/4$ or $3/4$. The no-pole gamma entry is

$$\begin{aligned} G_\alpha(m,m) = & -\log\pi + \operatorname{Re}\psi(\alpha - i\omega_m/2) + \frac{\operatorname{Re}\psi'(\alpha - i\omega_m/2)}{2S} \\ & - \frac{2}{S}\sum_{k\ge 0} e^{-b_k S}\frac{b_k^2 - \omega_m^2}{(b_k^2+\omega_m^2)^2}, \end{aligned} \tag{25}$$

$$\begin{aligned} G_\alpha(m,n) = & -\frac{\operatorname{Im}\psi(\alpha - i\omega_m/2) - \operatorname{Im}\psi(\alpha - i\omega_n/2)}{2\pi(m-n)} \\ & - \frac{1}{\pi(m-n)}\sum_{k\ge 0} e^{-b_k S}\left(\frac{\omega_m}{b_k^2+\omega_m^2} - \frac{\omega_n}{b_k^2+\omega_n^2}\right) \quad (m \ne n). \end{aligned} \tag{26}$$

They follow by expanding the gamma kernel and integrating the shifted Fourier correlations. Truncation before $k = K$ has a uniform absolute error

$$\varepsilon_{\alpha,K} \le \frac{2e^{-(2K+2\alpha)S}}{S(2K+2\alpha)^2(1-e^{-2S})}. \tag{27}$$

For the character controls use $G_{3/4} + \log(q)I$ and subtract the finite log-derivative correlation sum. For the Epstein controls use $G_{1/4} + G_{3/4} + \log(4d)I$, the one pole block, and that control's log-derivative correlation sum. Selected entries are checked against adaptive integration of the original regularized geometric kernel. The parity matrices also obey the exact divided-difference formulas implemented in the generator.

For completeness, the genus identities (4)–(5) can be obtained from the two reduced ideal classes. Their forms are $(1,0,5),(2,2,3)$ at discriminant -20, and $(1,0,6),(2,0,3)$ at discriminant -24. Decompose the principal partial zeta function into the trivial class character and the nontrivial genus character. The latter Euler products factor as $L(\chi_{-4})L(\chi_5)$ and $L(\chi_{-3})L(\chi_8)$, respectively. The unit group has order two, giving the representation normalization $E/2$.

The additional contours enclose all four sides, including the real bottom; they do not assume the classical DH positive-theta lemma for another control. FLINT's direct series evaluations on nonzero-radius cells containing removable points can be indeterminate. We use analytic Taylor images there. For a character combination with zero period sum, partial summation gives $|H(s)| \le |s| \sum_{n<q} |a_n| / \operatorname{Re} s$ for $\operatorname{Re} s > 0$. Thus $M = 3\sum_{n<q} |a_n|$ bounds $|H|$ on $|s-1| \le 1/2$. The Taylor remainder through order K on $|s-1| \le r < 1/2$ is at most

$$M \, \frac{(2r)^{K+1}}{1-2r}. \tag{28}$$

For $F_d = (s-1)H_d$, fractional-part partial summation gives $|(s-1)\zeta(s)| \le 3$ on the same disk. If M_j are the three character-period absolute sums in the genus decomposition, a valid bound is $(9M_0 + (9/2)M_1M_2)/2$. Near zero the regular functional-equation identity is

$$F_d(s) = \frac{\Gamma(2-s)}{\Gamma(1+s)} (\sqrt{d}/\pi)^{1-2s} F_d(1-s). \tag{29}$$

Cauchy bounds on a slightly larger Taylor disk also enclose the required derivative coefficients. These images, ordinary Taylor images elsewhere, and the Rouché inequalities are the numerical inputs to the finite zero counts.

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